

Problem 3

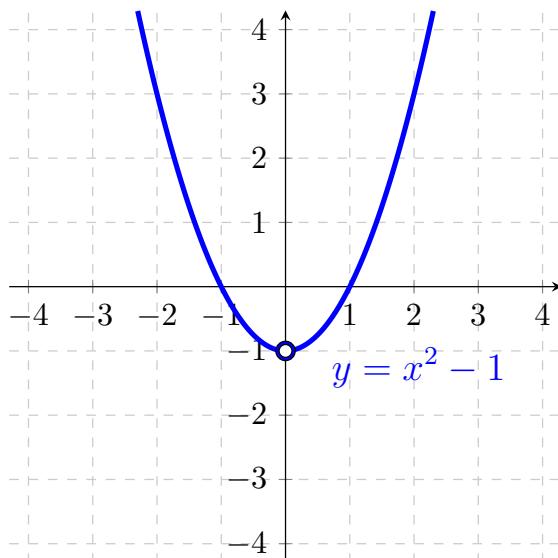
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Problem 1 Define $f(x) = \begin{cases} x^2 - 1 & \text{if } x < 0 \\ ? & \text{if } x > 0 \end{cases}$

(a) Find an expression for "?" such that f will be even.

Explanation. If f is even then $f(x) = f(-x)$.

$$\begin{aligned} f(x) &= f(-x) \text{ for } x > 0 \\ &= (-x)^2 - 1 \\ &= x^2 - 1 \end{aligned}$$



(b) Find an expression for "?" such that f will be odd.

Explanation. If f is odd then $f(-x) = -f(x)$.

$$\begin{aligned} -f(x) &= f(-x) \text{ for } x > 0 \\ &= (-x)^2 - 1 \\ &= x^2 - 1 \\ \implies f(x) &= -x^2 + 1 \end{aligned}$$

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